

Template Week 6 – Networking

Student number: 530010

Assignment 6.1: Working from home

Screenshot installation openssh-server:

```
ubuntu@ubuntu2404:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  openssh-client openssh-sftp-server
Suggested packages:
  keychain libpam-ssh monkeysphere ssh-askpass molly-guard
The following packages will be upgraded:
  openssh-client openssh-server openssh-sftp-server
3 upgraded, 0 newly installed, 0 to remove and 426 not upgraded.
Need to get 1,453 kB of archives.
After this operation, 0 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-sftp-server amd64 1:9.6p1-3ubuntu13.14 [37.3 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-server amd64 1:9.6p1-3ubuntu13.14 [510 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-client amd64 1:9.6p1-3ubuntu13.14 [906 kB]
Fetched 1,453 kB in 1s (1,211 kB/s)
Preconfiguring packages ...
(Reading database ... 196122 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-sftp-server (1:9.6p1-3ubuntu13.14) over (1:9.6p1-3ubuntu13.11) ...
Preparing to unpack .../openssh-server_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) over (1:9.6p1-3ubuntu13.11) ...
Preparing to unpack .../openssh-client_1%3a9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-client (1:9.6p1-3ubuntu13.14) over (1:9.6p1-3ubuntu13.11) ...
Setting up openssh-client (1:9.6p1-3ubuntu13.14) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Replacing config file /etc/ssh/sshd_config with new version
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
ubuntu@ubuntu2404:~$
```

Screenshot successful SSH command execution:

```
C:\Users\diede>ssh ubuntu@192.168.139.129
ubuntu@192.168.139.129's password:
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

407 updates can be applied immediately.
94 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Fri Oct 3 06:13:43 2025 from 192.168.139.1
ubuntu@ubuntu2404:~$
```

Screenshot successful execution SCP command:

```
C:\>scp info.txt ubuntu@192.168.139.129:/home/ubuntu
ubuntu@192.168.139.129's password:
info.txt 100% 36 35.2KB/s 00:00

C:\>ssh ubuntu@192.168.139.129
ubuntu@192.168.139.129's password:
Welcome to Ubuntu 24.04 LTS (GNU/Linux 6.14.0-33-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

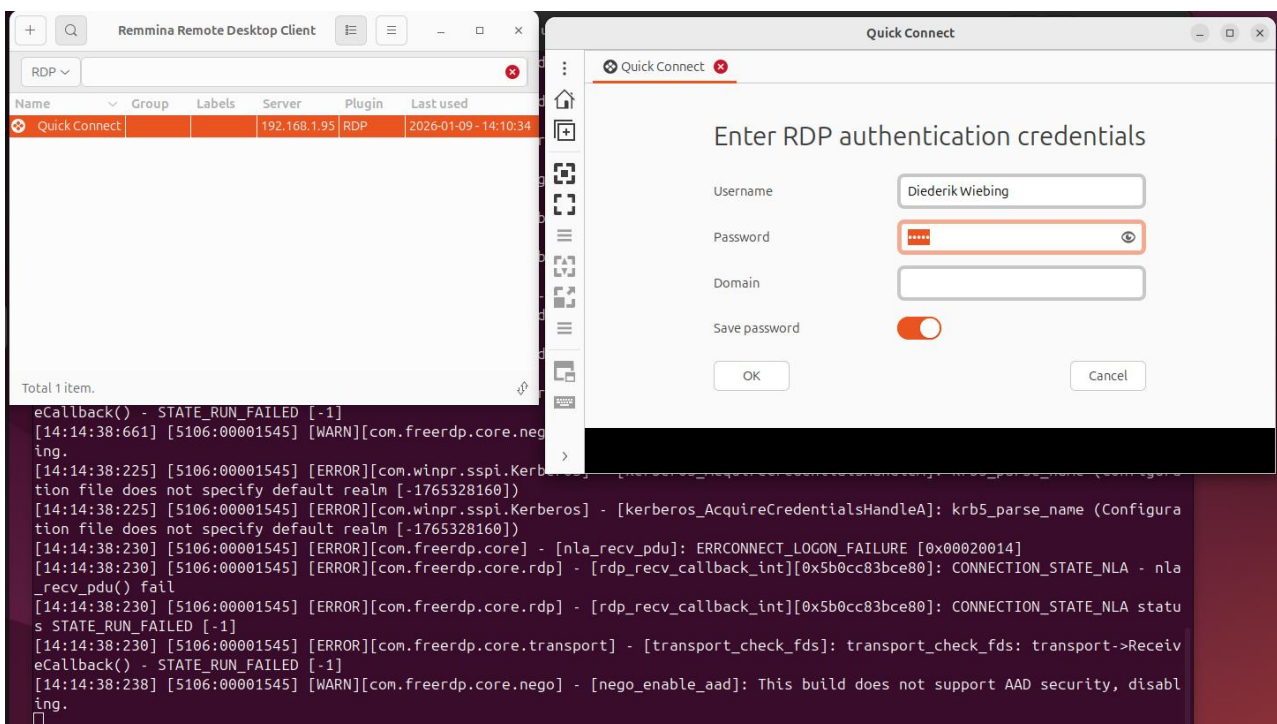
Expanded Security Maintenance for Applications is not enabled.

404 updates can be applied immediately.
94 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

6 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

1 updates could not be installed automatically. For more details,
see /var/log/unattended-upgrades/unattended-upgrades.log
Last login: Fri Jan 9 13:36:58 2026 from 192.168.139.1
ubuntu@ubuntu2404:~$ ls /home/ubuntu
Desktop      Documents  hello      Music      SherlockHolmes.txt  test
Diederik_530010.tar.gz  Downloads  image.gif  Pictures   snap               Videos
Diederik_530010.txt    email.txt  info.txt   Public     Templates
ubuntu@ubuntu2404:~$
```

Screenshot remmina:



Assignment 6.2: IP addresses websites

Relevant screenshots nslookup command:

```
ubuntu@ubuntu2404:~$ nslookup
> amazon.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   amazon.com
Address: 98.87.170.71
Name:   amazon.com
Address: 98.87.170.74
Name:   amazon.com
Address: 98.82.161.185
> google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   google.com
Address: 172.217.23.206
Name:   google.com
Address: 2a00:1450:400e:804::200e
>

> one.one.one.one
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   one.one.one.one
Address: 1.1.1.1
Name:   one.one.one.one
Address: 1.0.0.1
Name:   one.one.one.one
Address: 2606:4700:4700::1111
Name:   one.one.one.one
Address: 2606:4700:4700::1001
> dns.google.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   dns.google.com
Address: 8.8.8.8
Name:   dns.google.com
Address: 8.8.4.4
Name:   dns.google.com
Address: 2001:4860:4860::8844
>

> bol.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   bol.com
Address: 79.170.100.62
> w3scholls.com
Server:          127.0.0.53
Address:         127.0.0.53#53

Non-authoritative answer:
Name:   w3scholls.com
Address: 208.91.196.105
>
```

Screenshot website visit via IP address:



Assignment 6.3: subnetting

How many IP addresses are in this network configuration 192.168.110.128/25?

128 IP addresses in total

What is the usable IP range to hand out to the connected computers?

The usable IP range is 192.168.110.129 to 192.168.11.254

Check your two previous answers with this Linux command: `ipcalc 192.168.110.128/25`

```
ubuntu@ubuntu2404:~$ ipcalc 192.168.110.128/25
Address: 192.168.110.128      11000000.10101000.01101110.1 0000000
Netmask: 255.255.255.128 = 25 11111111.11111111.11111111.1 00000000
Wildcard: 0.0.0.127          00000000.00000000.00000000.0 11111111
=>
Network: 192.168.110.128/25   11000000.10101000.01101110.1 00000000
HostMin: 192.168.110.129      11000000.10101000.01101110.1 00000001
HostMax: 192.168.110.254      11000000.10101000.01101110.1 11111110
Broadcast: 192.168.110.255     11000000.10101000.01101110.1 11111111
Hosts/Net: 126                 Class C, Private Internet

ubuntu@ubuntu2404:~$
```

Explain the above calculation in your own words.

The /25 means the first 25 bits identify the network, leaving 7 bits for host addresses.

$2^7 = 128$ gives the total number of IPs in the subnet.

The first IP (network address) and the last IP (broadcast address) cannot be assigned to computers.

Therefore, the usable range for devices is 192.168.110.129 – 192.168.110.254.

Tools like ipcalc confirm this automatically and show network info clearly.

Assignment 6.4: HTML

Screenshot IP address Ubuntu VM:

```
ubuntu@ubuntu2404:~/site$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:3c:f2:6e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.139.129/24 brd 192.168.139.255 scope global dynamic noprefixroute ens33
        valid_lft 1764sec preferred_lft 1764sec
```

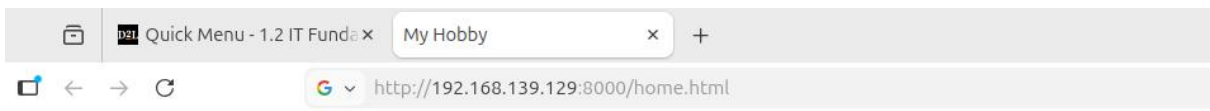
Screenshot of Site directory contents:

```
ubuntu@ubuntu2404:~/site$ ls
css          images      pdf         week2.html  week4.html  week6.html
home.html   index.html week1.html  week3.html  week5.html  week7.html
```

Screenshot python3 webserver command:

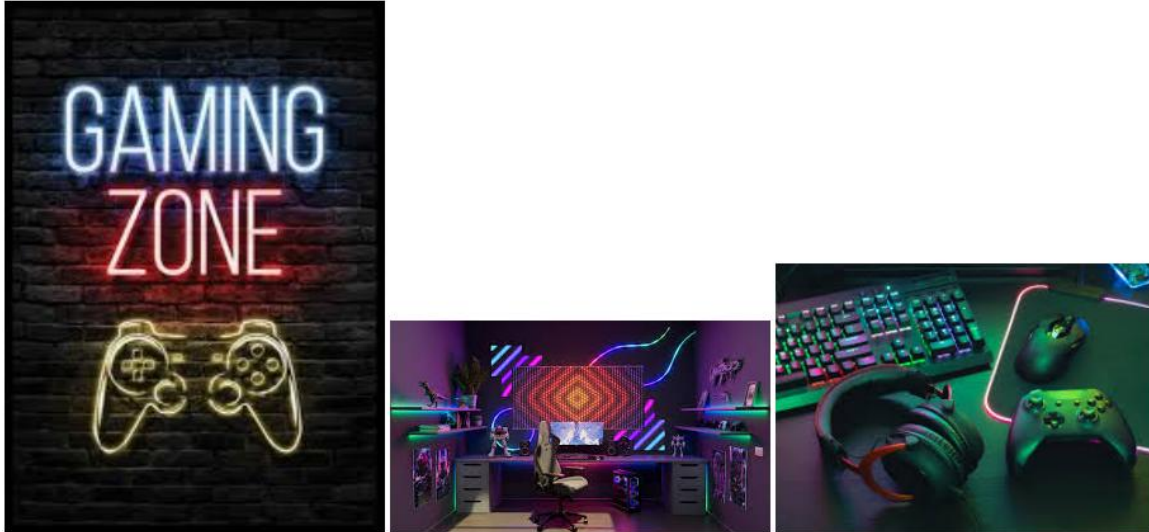
```
ubuntu@ubuntu2404:~/site$ python3 -m http.server 8000
Serving HTTP on 0.0.0.0 port 8000 (http://0.0.0.0:8000/) ...
192.168.139.1 - - [09/Jan/2026 15:24:37] "GET /home.html HTTP/1.1" 200 -
192.168.139.1 - - [09/Jan/2026 15:24:38] code 404, message File not found
192.168.139.1 - - [09/Jan/2026 15:24:38] "GET /home/ubuntu/site/images/gaming1.jpeg HTTP/1.1" 404 -
192.168.139.1 - - [09/Jan/2026 15:24:38] code 404, message File not found
192.168.139.1 - - [09/Jan/2026 15:24:38] code 404, message File not found
192.168.139.1 - - [09/Jan/2026 15:24:38] "GET /home/ubuntu/site/images/gaming3.jpeg HTTP/1.1" 404 -
192.168.139.1 - - [09/Jan/2026 15:24:38] "GET /home/ubuntu/site/images/gaming2.jpeg HTTP/1.1" 404 -
192.168.139.1 - - [09/Jan/2026 15:24:38] code 404, message File not found
192.168.139.1 - - [09/Jan/2026 15:24:38] "GET /favicon.ico HTTP/1.1" 404 -
```


Screenshot web browser visits your site



My Hobby: Photography

I love gaming and get angry at the game and at ryan, cuz he a bitchboy.



Assignment 6.5: Network segment

Remember that bitwise java application you've made in week 2? Expand that application so that you can also calculate a network segment as explained in the PowerPoint slides of week 6. Use the bitwise & AND operator. You need to be able to input two Strings. An IP address and a subnet.

IP: 192.168.1.100 and subnet: 255.255.255.224 for /27

Example: 192.168.1.100/27

Calculate the network segment

IP Address: 11000000.10101000.00000001.01100100

Subnet Mask: 11111111.11111111.11111111.11100000

Network Addr: 11000000.10101000.00000001.01100000

This gives 192.168.1.96 in decimal as the network address.

For a /27 subnet, each segment (or subnet) has 32 IP addresses (2^5).

The range of this network segment is from 192.168.1.96 to 192.168.1.127.

Paste source code here, with a screenshot of a working application.



The screenshot shows an IDE window with a Java file named Main.java. The code defines a public class NetworkCalculator with a static method ipToInt. A red squiggly line under the class name indicates a compilation error. The error message in the console on the right states: 'Main.java:3: error: class NetworkCalculator is public, should be declared in a public class NetworkCalculator {'. The error points to the opening curly brace of the class definition.

```
import java.util.Scanner;

public class NetworkCalculator {
    // Convert decimal IP string (e.g., "192.168.1.100") to 32-bit int
    public static int ipToInt(String ip) {
        import java.util.Scanner;

        public class NetworkCalculator {

            public static int ipToInt(String ip) {

                String[] parts = ip.split("\\.");

                int result = 0;

                for (int i = 0; i < 4; i++) {

                    result = (result << 8) | Integer.parseInt(parts[i]);

                }

                return result;

            }

            public static String intToIp(int ipInt) {

                return ((ipInt >> 24) & 0xFF) + "." +

                    ((ipInt >> 16) & 0xFF) + "." +

                    ((ipInt >> 8) & 0xFF) + "." +

                    (ipInt & 0xFF);

            }

            public static String ipToBinaryString(int ipInt) {

                return String.format("%8s", Integer.toBinaryString((ipInt >> 24) & 0xFF)).replace(' ', '0') + "." +

                    String.format("%8s", Integer.toBinaryString((ipInt >> 16) & 0xFF)).replace(' ', '0') + "." +

                    String.format("%8s", Integer.toBinaryString((ipInt >> 8) & 0xFF)).replace(' ', '0') + "." +

                    String.format("%8s", Integer.toBinaryString(ipInt & 0xFF)).replace(' ', '0');

            }

        }
    }
}
```

Main.java:3: error: class NetworkCalculator is public, should be declared in a public class NetworkCalculator {
^
1 error

```
public class NetworkCalculator {
```

```
    public static int ipToInt(String ip) {
```

```
        String[] parts = ip.split("\\.");
```

```
        int result = 0;
```

```
        for (int i = 0; i < 4; i++) {
```

```
            result = (result << 8) | Integer.parseInt(parts[i]);
```

```
        }
```

```
        return result;
```

```
    }
```

```
    public static String intToIp(int ipInt) {
```

```
        return ((ipInt >> 24) & 0xFF) + "." +
```

```
            ((ipInt >> 16) & 0xFF) + "." +
```

```
            ((ipInt >> 8) & 0xFF) + "." +
```

```
            (ipInt & 0xFF);
```

```
    }
```

```
    public static String ipToBinaryString(int ipInt) {
```

```
        return String.format("%8s", Integer.toBinaryString((ipInt >> 24) & 0xFF)).replace(' ', '0') + "." +
```

```
            String.format("%8s", Integer.toBinaryString((ipInt >> 16) & 0xFF)).replace(' ', '0') + "." +
```

```
            String.format("%8s", Integer.toBinaryString((ipInt >> 8) & 0xFF)).replace(' ', '0') + "." +
```

```
            String.format("%8s", Integer.toBinaryString(ipInt & 0xFF)).replace(' ', '0');
```

```
    }
```

```

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter IP address (e.g., 192.168.1.100): ");
    String ipStr = sc.nextLine();

    System.out.print("Enter subnet mask (e.g., 255.255.255.224): ");
    String subnetStr = sc.nextLine();

    int ip = ipToInt(ipStr);
    int subnet = ipToInt(subnetStr);

    int network = ip & subnet;

    int hostBits = 32 - Integer.bitCount(subnet);
    int numHosts = (int) Math.pow(2, hostBits);

    int broadcast = network | (~subnet);

    System.out.println("\nIP Address:   " + ipToBinaryString(ip));
    System.out.println("Subnet Mask:  " + ipToBinaryString(subnet));
    System.out.println("-----");
    System.out.println("Network Addr: " + ipToBinaryString(network));
    System.out.println("Network Addr (decimal): " + intToIp(network));
    System.out.println("Number of IPs in this subnet: " + numHosts);
    System.out.println("Usable IP range: " + intToIp(network) + " - " + intToIp(broadcast));
}
}

```

Ready? Save this file and export it as a pdf file with the name: [week6.pdf](#)